Simplex method

Algorithm.

Step 1

Z maximize

Step 2

In equalities → Equation

we add Slack variable

Step 3

Initial simplex Tableau.

Step 4 pivot colum selecte

Step 5 min Ratio Test

Step 6 Repeat

unit no negative value occur

Step 7 solution read.

Solving standard maximization problem using the simplex method.

maximize ~~Minimize~~ $Z = 8x_1 + 10x_2 + 7x_3$

Subject to

$$x_1 + 3x_2 + 2x_3 \leq 10$$

$$x_1 + 5x_2 + x_3 \leq 8$$

$$x_1, x_2, x_3 > 0$$

Solution

2 Slack variable

3 variable

$$x_1 + 3x_2 + 2x_3 + S_1 + 0.S_2 = 10$$

$$x_1 + 5x_2 + x_3 + 0.S_1 + 1S_2 = 8$$

$$-8x_1 - 10x_2 - 7x_3 + Z = 0$$

x_1	x_2	x_3	S_1	S_2	Z	
1	3	2	1	0	0	10
1	5	1	0	1	0	8
-8	-10	-7	0	0	1	0

most Negative Indicator

X_1	X_2	X_3	S_1	S_2	Z	
1	3	2	1	0	0	10
1	5	1	0	1	0	8
-8	-10	-7	0	0	1	0

$10/3 = 3\,1/3$

$8/5 = 1\,3/5$

$1/5 R_2 \rightarrow R_2$

$-3R_2 + R_1 \rightarrow R_1$

$10R_2 + R_3 \rightarrow R_3$

X_1	X_2	X_3	S_1	S_2	Z	
2/5	0	7/5	1	3/5	0	26/5
1/5	1	1/5	0	1/5	0	8/5
-6	0	-5	0	2	1	16

most negative Indicator

X_1	X_2	X_3	S_1	S_2	Z	
2/5	0	7/5	1	3/5	0	26/5
1/5	1	1/5	0	1/5	0	8/5
-6	0	-5	0	2	1	16

$26/5 \div 2/5 = 1$

$8/5 \div 1/5 = 8$

$5R_2 \rightarrow R_2$

$-2/5 R_2 + R_1 \rightarrow R_1$

$6R_2 + R_3 \rightarrow R_3$

x_1	x_2	x_3	s_1	s_2	z	
0	-2	1	1	-1	0	2
1	5	1	0	1	0	8
0	30	1	0	8	1	64

no negative entries
optimal solution

$x_1 = 8 \quad x_2 = 0 \quad x_3 = 0 \quad s_1 = 2 \quad s_2 = 0 \quad z = 64$

maximum value of 64 at $(8, 0, 0)$